

Cyber Security Internship – Task 2

Phishing Email Detection & Awareness System

Cyber Security Internship – Task 2
Future Interns

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1. Introduction

Phishing is one of the most common forms of cyberattacks used by threat actors to steal sensitive information such as usernames, passwords, banking credentials, and personal data. These attacks typically rely on social engineering techniques to manipulate users into clicking malicious links, downloading infected attachments, or sharing confidential information.

Modern phishing attacks often imitate trusted organizations such as banks, email providers, delivery services, or corporate departments to appear legitimate. Attackers use urgency, fear, curiosity, or financial rewards to increase the likelihood of user interaction.

The purpose of this project is to analyze phishing email samples using practical cybersecurity investigation techniques. The report focuses on identifying phishing indicators, analyzing suspicious domains and email content, classifying threats, and creating awareness guidelines that help users recognize and avoid phishing attacks.

This project was conducted as part of the Cyber Security Internship program at [Future Interns](#) and is based on practical phishing analysis scenarios performed in a controlled learning environment.

2. Objectives

The primary objectives of this project are:

- To analyze suspicious phishing email samples
- To identify common phishing indicators and social engineering techniques
- To investigate suspicious sender domains and URLs
- To classify phishing threats based on risk level
- To understand attacker tactics used in phishing campaigns
- To document findings in a professional incident-report format
- To create awareness and prevention guidelines for users
- To improve practical SOC analyst and threat investigation skills

3. Tools & Technologies Used

The following tools and platforms were used during the phishing investigation process:

Tool	Purpose
TryHackMe	Practical phishing investigation labs
VirusTotal	Domain and URL reputation analysis
MXToolbox	Email header analysis
Google Header Analyzer	SPF/DKIM email verification
Browser Developer Tools	URL and link inspection
GitHub	Project hosting and documentation
MS Word / Google Docs	Report preparation

These tools helped identify phishing indicators, suspicious domains, email authentication failures, and malicious behaviors associated with phishing campaigns.

4. Methodology

The phishing investigation process followed a structured workflow similar to procedures used in Security Operations Centers (SOCs).

Step 1 – Collection of Phishing Samples

Phishing email samples were collected from controlled cybersecurity training environments and phishing simulation labs.

Step 2 – Email Content Analysis

Each email was reviewed to identify suspicious characteristics such as:

- urgency or fear tactics
- financial lure techniques
- suspicious sender information
- requests for sensitive information

- suspicious URLs or attachments

Step 3 – Indicator Investigation

Technical indicators were investigated, including:

- sender domains
- embedded URLs
- email headers
- SPF/DKIM authentication results
- reputation analysis using VirusTotal

Step 4 – Risk Classification

Each email was classified into one of the following categories:

- Safe
- Suspicious
- Phishing

Risk levels were assigned based on the severity of identified indicators.

Step 5 – Documentation & Reporting

Findings, screenshots, attack indicators, and remediation recommendations were documented in a professional cybersecurity incident-report format.

5. Phishing Fundamentals

Phishing is a social engineering attack designed to trick users into performing actions that compromise security. These attacks commonly target employees, businesses, and online users.

Common Types of Phishing

Credential Harvesting

Attackers create fake login pages to steal usernames and passwords.

Financial Scam Phishing

Victims are tricked into sharing banking information or sending money.

Attachment-Based Phishing

Malicious files such as PDFs, ZIP files, or Office documents are delivered through email.

Business Email Compromise (BEC)

Attackers impersonate executives or trusted employees to manipulate financial transactions.

Brand Impersonation

Cybercriminals imitate trusted organizations such as banks, Microsoft, PayPal, Amazon, or delivery companies.

Common Social Engineering Techniques

Attackers often use:

- urgency
- fear
- authority
- curiosity
- financial rewards

to manipulate users into acting quickly without verification.

6. Case Study 1 – Financial Scam Phishing Email

Assigned alert(s)

Write case report

1000

Suspicious email from external domain.

Low

Phishing

May 25th 2026 at 17:22

Description:

A suspicious email was received from an external sender with an unusual top level domain. Note from SOC Lead: This detection rule still needs fine-tuning.

datasource:

email

timestamp:

05/25/2026 17:20:57.634

subject:

Inheritance Alert: Unknown Billionaire Relative Left You Their Hat Fortunes

sender:

eileen@trendymillineryco.me

recipient:

support@tryhatme.com

attachment:

None

content:

A long lost billionaire relative has left you their secret hat empire To claim your inheritance send us your banking details immediately

direction:

inbound

Alert Information

Field **Value**

Alert ID 1000

Alert Type Phishing

Severity Low

Direction Inbound

Email Details

Field **Value**

Subject Inheritance Alert: Unknown Billionaire Relative Left You Their Hat Fortunes

Sender eileen@trendymillineryco.me

Recipient support@tryhackme.com

Analysis

The email attempts to lure the victim using a fake inheritance claim. The attacker uses emotional and financial manipulation to convince the user to share banking details.

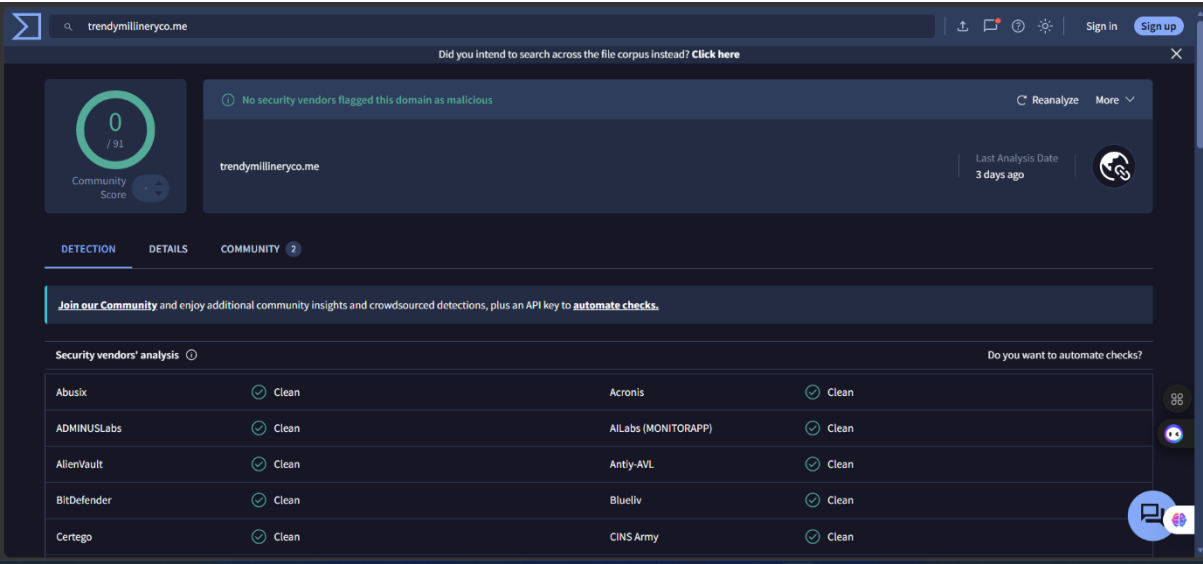
The message contains multiple phishing indicators:

- unrealistic financial claims
- suspicious sender domain
- request for sensitive banking information

- social engineering tactics

Domain Investigation

The sender domain was analyzed using VirusTotal. Although the domain was not flagged as malicious, this does not guarantee legitimacy. Newly created or low-reputation phishing domains may not yet be detected by security vendors.



Risk Classification

Category Result

Threat Type Phishing

Risk Level Medium

INCIDENT REPORT

Time of Activity:

05/25/2026 17:20:57 UTC

List of Affected Entities:

- Recipient: support@tryhatme.com
 - Sender Domain: trendymillineryco.me
 - Email Account/User targeted by phishing campaign
-

Reason for Classifying as True Positive:

The alert was classified as a true positive because the email contains multiple phishing indicators, including suspicious sender information, unrealistic inheritance claims, social engineering tactics, and requests for sensitive banking information.

The message attempts to manipulate the recipient into disclosing confidential financial details using emotional and financial lure techniques commonly observed in phishing campaigns.

Reason for Escalating the Alert:

The alert requires escalation because the email demonstrates malicious phishing behavior that could result in credential theft, financial fraud, or unauthorized disclosure of sensitive information if interacted with by the recipient.

Further investigation and blocking actions are recommended to prevent additional phishing attempts from the same domain or sender.

Recommended Remediation Actions:

- Block the sender domain: trendymillineryco.me
- Quarantine similar phishing emails
- Educate users about inheritance and financial scam emails
- Monitor for additional emails from related domains
- Add the domain to email filtering and threat intelligence systems

- Advise users not to share banking or personal information through email
-

List of Attack Indicators:

- Suspicious sender domain
- Financial lure/social engineering tactics
- Request for banking information
- Unrealistic inheritance claim
- External email source
- Urgency and manipulation language
- Unknown sender identity

7. Case Study 2 – HR Onboarding Phishing Email

Assigned alert(s)

Write case report

8814	Inbound Email Containing Suspicious External Link	^	Medium	Phishing	May 26th 2026 at 14:19	
Description:		This alert was triggered by an inbound email contains one or more external links due to potentially suspicious characteristics. As part of the investigation, check firewall or proxy logs to determine whether any endpoints have attempted to access the URLs in the email and whether those connections were allowed or blocked.				
datasource:		email				
timestamp:		05/26/2026 14:17:41.824				
subject:		Action Required: Finalize Your Onboarding Profile				
sender:		onboarding@hrconnex.thm				
recipient:		j.garcia@thetrydaily.thm				
attachment:		None				
content:		Hi Ms. Garcia,\n\nWelcome to TheTryDaily!\n\nAs part of your onboarding, please complete your final profile setup so we can configure your access.\n\nKindly please click the link below:\n\n https://hrconnex.thm/onboarding/15400654060/j.garcia >Set Up My Profile\n\nIf you have questions, please reach out to the HR Onboarding Team.				
direction:		inbound				

Alert Information

Field Value

Alert ID 8814

Alert Type Phishing

Severity Medium

Direction Inbound

Email Details

Field Value

Subject Action Required: Finalize Your Onboarding Profile

Sender onboarding@hrconnex.thm

Recipient j.garcia@thetrydaily.thm

Analysis

The email attempted to impersonate an HR onboarding process to trick the recipient into clicking a suspicious external link. The attacker used urgency and onboarding-related language to create legitimacy and encourage user interaction.

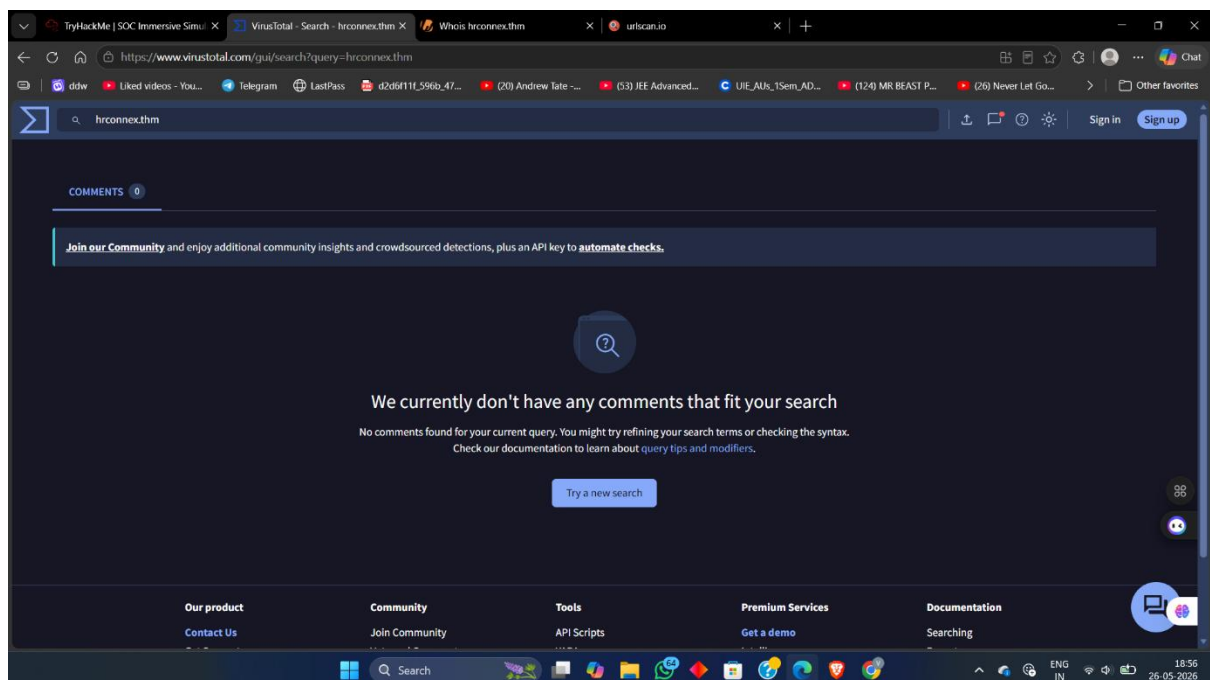
The message contained multiple phishing indicators:

- suspicious external onboarding link
- impersonation of HR onboarding services
- urgent action request
- suspicious/non-legitimate sender domain
- possible credential harvesting attempt
- social engineering tactics

The embedded link directed the user to:

<https://hrconnex.thm/onboarding/15400654060/j.garcia>

The email attempted to convince the recipient to complete a profile setup process, which is a common phishing technique used to steal credentials or sensitive information.



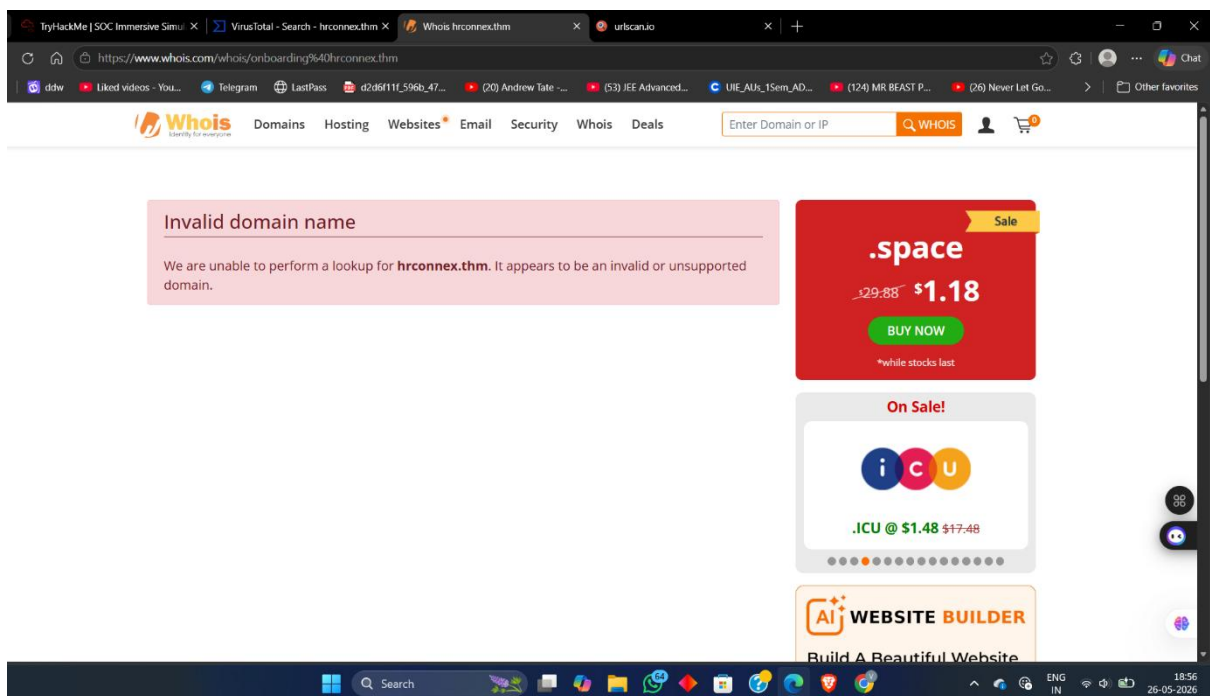
Domain Investigation

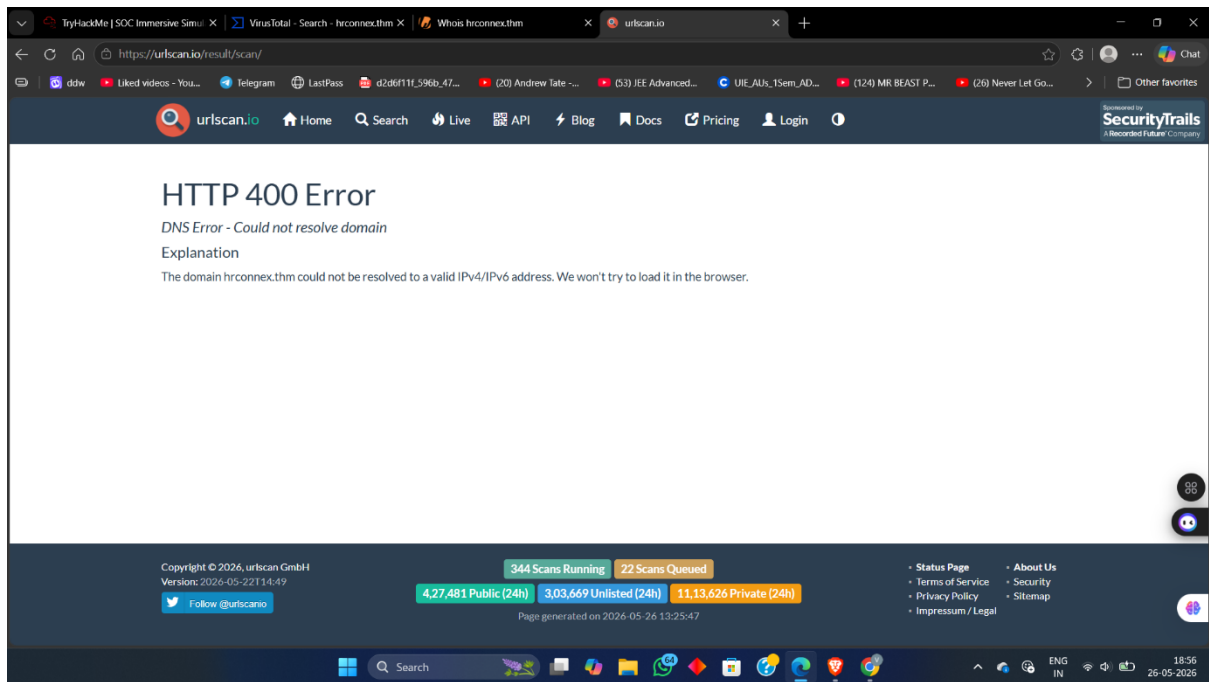
The sender domain hrconnex.thm was investigated using VirusTotal and URLScan.

Findings included:

- VirusTotal returned no valid reputation information for the domain.
- URLScan produced a DNS resolution failure.
- The domain could not resolve to a valid IPv4/IPv6 address.
- WHOIS lookup identified the domain as invalid/unregistered.

Although the domain was not actively flagged as malicious, the lack of legitimate infrastructure and DNS resolution strongly indicated suspicious phishing activity.





Risk Classification

Category Result

Threat Type Phishing

Risk Level Medium

INCIDENT REPORT

Time of Activity:

05/26/2026 14:17:41 UTC

List of Affected Entities:

- Recipient: j.garcia@thetrydaily.thm
 - Suspicious Sender: onboarding@hrconnex.thm
 - Suspicious Domain: hrconnex.thm
 - Email Subject: Action Required: Finalize Your Onboarding Profile
-

Reason for Classifying as True Positive:

The alert was classified as a True Positive phishing incident due to multiple phishing indicators identified during investigation. The email used social engineering techniques by creating urgency and requesting the recipient to complete onboarding actions through an external link. The sender domain hrconnex.thm appeared suspicious and could not be resolved through WHOIS or DNS lookup checks. URLScan returned a DNS resolution failure, and no legitimate reputation information was found on VirusTotal. These findings indicate a suspicious and potentially malicious phishing attempt.

Reason for Escalating the Alert:

The alert was escalated because the phishing email targeted a user with a credential/profile setup lure that could potentially lead to credential theft or unauthorized access. The use of an external suspicious link and impersonation of an HR onboarding process increased the risk level and required further review by higher-level security personnel.

Recommended Remediation Actions:

- Block the domain hrconnex.thm at the email gateway and firewall level.
- Quarantine similar phishing emails identified in the environment.
- Inform the affected user about the phishing attempt.
- Advise the user not to click suspicious onboarding or credential-related links.
- Conduct additional monitoring for any attempted access related to the suspicious domain.

- Add the identified indicators to SIEM/watchlists for future detection.
-

List of Attack Indicators:

- Suspicious sender address: onboarding@hrconnex.thm
- Suspicious external domain: hrconnex.thm
- External onboarding URL embedded in email
- Urgent call-to-action language
- HR impersonation/social engineering tactics
- DNS resolution failure for the domain
- Invalid WHOIS lookup results
- No trusted reputation data available on VirusTotal

8. Case Study 3 – Blacklisted External URL Access Attempt

8816

Access to Blacklisted External URL Blocked by Firewall

^

High

Firewall

May 26th 2026 at 14:24

Description:

This alert was triggered when a user attempted to access an external URL that is listed in the organization's blacklist or threat intelligence feeds. The firewall or proxy successfully blocked the outbound request, preventing the connection. Note: The blacklist only covers known threats. It does not guarantee protection against new or unknown malicious domains.

datasource:

firewall

timestamp:

05/26/2026 14:22:08.824

Action:

blocked

SourceIP:

10.20.2.17

SourcePort:

34257

DestinationIP:

67.199.248.11

DestinationPort:

80

URL:

http://bit.ly/3sHkX3da12340

Application:

web-browsing

Protocol:

TCP

Rule:

Blocked Websites

Alert Information

Field	Value
-------	-------

Alert ID	8816
----------	------

Alert Type	Firewall
------------	----------

Severity	High
----------	------

Direction	Outbound
-----------	----------

Network Activity Details

Field	Value
-------	-------

Source IP	10.20.2.17
-----------	------------

Destination IP	67.199.248.11
----------------	---------------

URL	http://bit.ly/3sHkX3da12340
-----	-----------------------------

Action	Blocked
--------	---------

Protocol	TCP
----------	-----

Destination Port	80
------------------	----

Field	Value
Firewall Rule	Blocked Websites

Analysis

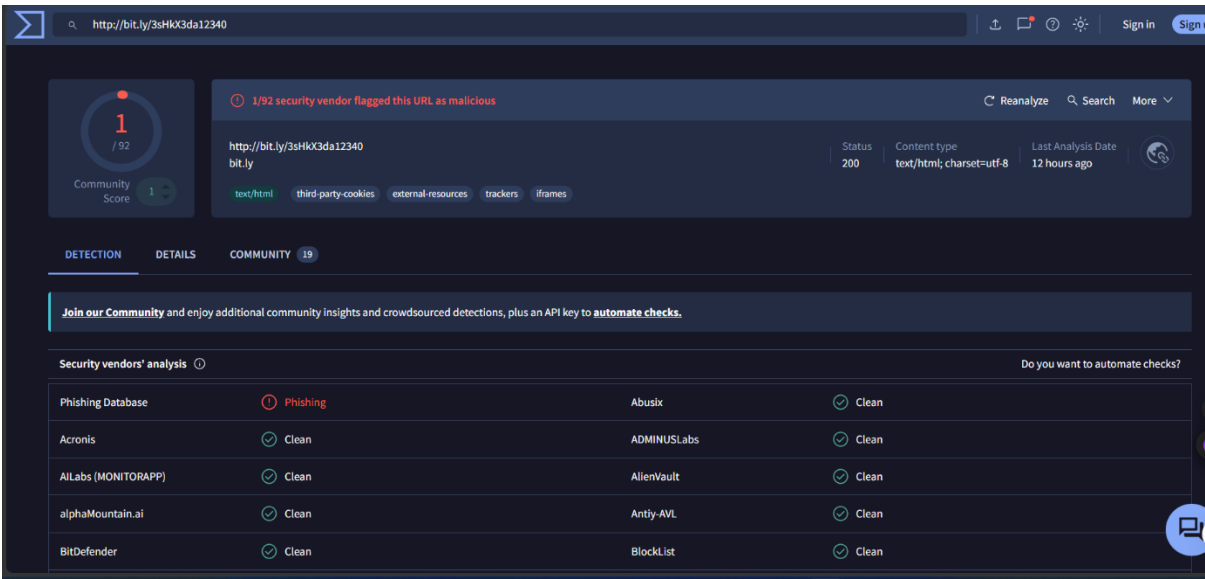
The alert was triggered when an internal host attempted to access a blacklisted external URL. The firewall successfully blocked the outbound connection attempt, preventing communication with the suspicious destination.

The URL used a shortened bit.ly link, which is commonly abused in phishing and malware campaigns to conceal malicious destinations and evade detection mechanisms.

The investigation identified multiple suspicious indicators:

- blacklisted external URL access attempt
- shortened URL usage (bit.ly)
- suspicious outbound web traffic
- firewall security policy trigger
- potential phishing or malware-related activity
- attempted communication with external infrastructure

The URL was analyzed using VirusTotal and URLScan.



URL Investigation

The suspicious URL was investigated to determine its reputation and redirection behavior.

Findings included:

- VirusTotal flagged the URL as phishing-related by one security vendor.
- URLScan analysis showed multiple redirects associated with the shortened link.
- The connection ultimately redirected through external infrastructure before reaching a legitimate site.
- The firewall blocked the traffic successfully before communication could be completed.

Although some security vendors marked the URL as clean, the blacklist trigger, shortened URL behavior, and phishing-related detections indicated suspicious activity consistent with malicious campaigns.

www.google.com

142.251.157.119 Public Scan

Submitted URL: <http://bit.ly/3sHkX3da12340> 13yr old

Effective URL: <https://www.google.com/> 56yr old

Submission: On May 26 via manual (May 26th 2026, 1:34:10 pm UTC) from IN — Scanned from PL

[Summary](#) [HTTP 34](#) [Redirects](#) [Behaviour](#) [Indicators](#) [Similar](#) [DOM](#) [Content](#) [API](#) [Verdicts](#)

Summary

This website contacted 7 IPs in 4 countries across 10 domains to perform 34 HTTP transactions. The main IP is 142.251.157.119, located in United States and belongs to GOOGLE - Google LLC, US. The main domain is www.google.com. The Cisco Umbrella rank of the primary domain is 3. 56yr old

TLS certificate: Issued by WE2 on May 7th 2026. Valid for: 3mo.

[bit.ly](#) scanned 10000+ times on urlscan.io [Show Scans 10000+](#)

[www.google.com](#) scanned 10000+ times on urlscan.io [Show Scans 10000+](#)

urlscan.io Verdict: No classification

Live information

Google Safe Browsing: No classification for www.google.com

Current DNS A record: 142.251.152.119 (AS15169 - GOOGLE - Google LLC, US)

Domain created: September 15th 1997, 12:30:00 (UTC)

Domain registrar: MarkMonitor, Inc.

Lookup

Go To

Rescan

Add Verdict

Report

Screenshot [Live screenshot](#) [Full Image](#)

Page Title

Google

Page URL History [Show full URLs](#)

[4](#) [https://www.google.com/](#) [https://www.google.com/](#) [https://www.google.com/](#) [https://www.google.com/](#)

Domain & IP information

IP/ASNs

IP Detail

Domains

Domain Tree

Links

Certs

Frames

	IP Address	AS Autonomous System
1 → 1	67.199.248.10	396982 (GOOGLE-CLOUD-PLATFORM - Google LLC)
3	188.114.97.3	13335 (CLOUDFLARENET - Cloudflare)
14	139.45.197.121	9002 (RETN-AS RETN Limited)
6 → 11	87.250.250.119	13238 (YANDEX YANDEX LLC)
2	172.64.146.234	13335 (CLOUDFLARENET - Cloudflare)
1	172.64.145.116	13335 (CLOUDFLARENET - Cloudflare)
2	142.251.157.119	15169 (GOOGLE - Google LLC)
34	7	

1. <http://bit.ly/3sHkX3da12340> HTTP 307

https://bit.ly/3sHkX3da12340 HTTP 301

https://rclcomlink.com/loading?n2026=3sHkX3da12340

Page URL

2. https://073m.com/4/3826381?var=short Page URL

3. https://www.google.com/ Page URL

Detected technologies

Yandex.Metrica (Analytics) Expand

Page Statistics

34	71 %	0 %	10	10
Requests	HTTPS	IPv6	Domains	Subdomains
7	4	119 kB	436 kB	25
IPs	Countries	Transfer	Size	Cookies



Type / to search

Products ▾

Data ▾

Why IPinfo? ▾

10.20.2.17

! Bogon IP Address

What is a Bogon IP Address?

Some IP addresses and IP ranges are reserved for special use, such as for local or private networks, and should not appear on the public internet. These reserved ranges, along with other IP ranges that haven't yet been allocated and therefore also shouldn't appear on the public internet are sometimes known as bogons.

Read more

Risk Classification

Category	Result
----------	--------

Threat Type	Suspicious External URL / Possible Phishing
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Risk Level	High
------------	------

INCIDENT REPORT

Time of Activity:

05/26/2026 14:22:08 UTC

List of Affected Entities:

- Source IP: 10.20.2.17
 - Destination IP: 67.199.248.11
 - URL Accessed: <http://bit.ly/3sHkX3da12340>
 - Firewall Rule Triggered: Blocked Websites
-

Reason for Classifying as True Positive:

The alert was classified as a True Positive because an internal host attempted to access a blacklisted external URL that triggered the organization's firewall protection mechanisms. Investigation revealed the use of a shortened URL (bit.ly), which is commonly abused in phishing and malicious campaigns to conceal final destinations. VirusTotal analysis identified the URL as phishing-related, and the firewall successfully blocked the outbound connection attempt. The activity matched known suspicious web-browsing behavior patterns.

Reason for Escalating the Alert:

The alert was escalated due to attempted communication with a blacklisted external URL from an internal system. Although the firewall blocked the connection successfully, the activity indicates potentially risky user behavior or possible phishing interaction that requires additional endpoint verification and monitoring.

Recommended Remediation Actions:

- Verify that the firewall fully blocked the outbound connection.
- Investigate the endpoint associated with source IP 10.20.2.17.
- Review browser history and recent user activity.
- Scan the host for malware or suspicious processes.
- Block the URL/domain across security controls if not already blocked.
- Educate the user regarding suspicious shortened URLs and phishing attempts.

- Continue monitoring for repeated access attempts.
-

List of Attack Indicators:

- Blacklisted external URL access attempt
- Shortened URL usage (bit.ly)
- VirusTotal phishing detection
- High severity firewall alert
- Suspicious outbound web request
- Firewall policy trigger: Blocked Websites
- External IP communication attempt on port 80/TCP

9. Case Study 4 – HR Onboarding Phishing Email

8818	Inbound Email Containing Suspicious External Link	Medium	Phishing	May 26th 2026 at 14:25	
Description:	This alert was triggered by an inbound email contains one or more external links due to potentially suspicious characteristics. As part of the investigation, check firewall or proxy logs to determine whether any endpoints have attempted to access the URLs in the email and whether those connections were allowed or blocked.				
datasource:	email				
timestamp:	05/26/2026 14:23:40.824				
subject:	Action Required: Finalize Your Onboarding Profile				
sender:	onboarding@hrconnex.thm				
recipient:	j.garcia@thetrydaily.thm				
attachment:	None				
content:	Hi Ms. Garcia,\n\nWelcome to TheTryDaily!\n\nAs part of your onboarding, please complete your final profile setup so we can configure your access.\n\nKindly click the link below:\n\nSet Up My Profile\n\nIf you have questions, please reach out to the HR Onboarding Team.				
direction:	inbound				

Alert Information

Field Value

Alert ID 8818

Alert Type Phishing

Severity Medium

Direction Inbound

Email Details

Field Value

Subject Action Required: Finalize Your Onboarding Profile

Sender onboarding@hrconnex.thm

Recipient j.garcia@thetrydaily.thm

Analysis

The email attempted to impersonate an HR onboarding process to trick the recipient into interacting with a suspicious external link. The message created urgency by requesting the user to complete profile setup actions required for onboarding.

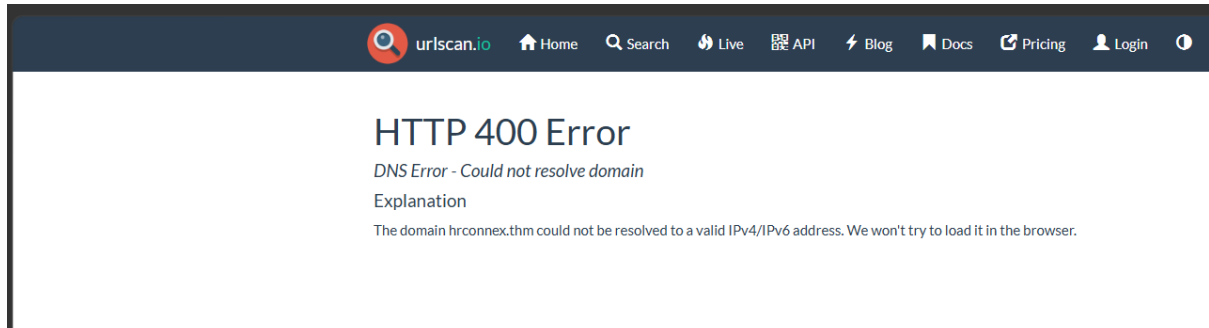
The email contained several phishing indicators:

- suspicious external onboarding link
- impersonation of HR onboarding services
- urgent action request
- suspicious sender domain
- possible credential harvesting attempt
- social engineering tactics

The embedded external link directed the recipient to:

<https://hrconnex.thm/onboarding/15400654060/j.garcia>

The attacker attempted to make the email appear legitimate by using professional onboarding language and requesting immediate action from the user.



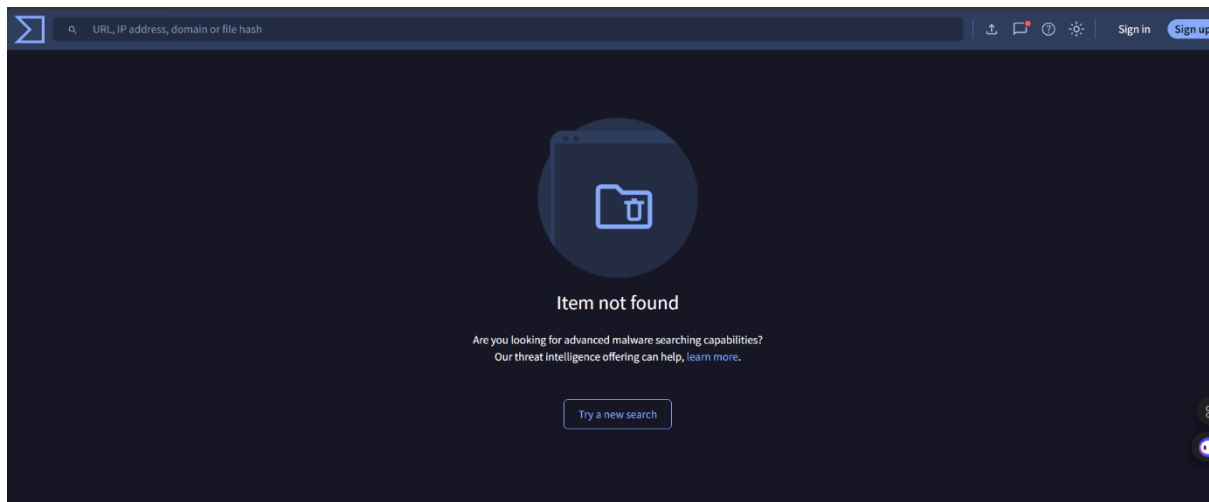
Domain Investigation

The domain hrconnex.thm was investigated using VirusTotal, URLScan, and WHOIS lookup tools.

Findings included:

- VirusTotal returned no valid reputation information for the domain.
- URLScan produced a DNS resolution failure.
- WHOIS lookup identified the domain as invalid/unregistered.
- The domain could not resolve to a valid IPv4/IPv6 address.

These findings strongly indicated suspicious or illegitimate infrastructure commonly associated with phishing campaigns and credential harvesting attempts.



Risk Classification

Category	Result
Threat Type	Phishing
Risk Level	Medium

INCIDENT REPORT

Time of Activity:

05/26/2026 14:23:40 UTC

List of Affected Entities:

- Recipient: j.garcia@thetrydaily.thm
 - Suspicious Sender: onboarding@hrconnex.thm
 - Suspicious Domain: hrconnex.thm
 - Email Subject: Action Required: Finalize Your Onboarding Profile
-

Reason for Classifying as True Positive:

The alert was classified as a True Positive phishing incident after investigation identified multiple phishing and social engineering indicators within the email. The message attempted to impersonate an HR onboarding process and urged the recipient to click an external link to finalize account setup. The embedded domain hrconnex.thm was investigated using VirusTotal and URLScan. VirusTotal returned no valid reputation information for the domain, while URLScan produced a DNS resolution failure, indicating that the domain could not resolve to a valid IP address. These findings suggest suspicious or illegitimate infrastructure commonly associated with phishing activity.

Reason for Escalating the Alert:

The alert was escalated because the email contained a suspicious external link potentially intended for credential harvesting or unauthorized access attempts. The phishing indicators, combined with the suspicious non-resolving domain and impersonation tactics, posed a security risk to the targeted user and organization.

Recommended Remediation Actions:

- Quarantine and remove the phishing email from affected mailboxes.
- Block the sender domain hrconnex.thm at the email gateway and firewall level.
- Search for similar phishing emails across the environment.
- Notify the affected user regarding the phishing attempt.
- Advise users not to interact with suspicious onboarding or credential-related links.

- Add the identified indicators to SIEM detection rules and blocklists.
 - Continue monitoring for additional suspicious email activity.
-

List of Attack Indicators:

- Suspicious sender address: onboarding@hrconnex.thm
- Suspicious domain: hrconnex.thm
- External onboarding link embedded in email
- HR impersonation/social engineering attempt
- Urgent call-to-action language
- DNS resolution failure identified in URLScan
- No valid reputation information in VirusTotal
- Potential credential harvesting behavior

10. Case Study 5 – Microsoft Account Phishing Email

8817	Inbound Email Containing Suspicious External Link	Medium	Phishing	May 26th 2026 at 14:25	
Description:		This alert was triggered by an inbound email contains one or more external links due to potentially suspicious characteristics. As part of the investigation, check firewall or proxy logs to determine whether any endpoints have attempted to access the URLs in the email and whether those connections were allowed or blocked.			
datasource:		email			
timestamp:		05/26/2026 14:23:12.824			
subject:		Unusual Sign-In Activity on Your Microsoft Account			
sender:		no-reply@microsoftsupport.co			
recipient:		c.allen@thetrydaily.thm			
attachment:		None			
content:		Hi C.Allen,\n\nWe detected an unusual sign-in attempt on your Microsoft account.\n\nLocation: Lagos, Nigeria\n\nIP Address: 102.89.222.143\n\nDate: 2025-01-24 06:42\n\nIf this was not you, please secure your account immediately to avoid unauthorized access.\n\n https://microsoftsupport.co/login >Review Activity\n\nThank you,\n\nMicrosoft Account Security Team			
direction:		inbound			

Alert Information

Field Value

Alert ID 8817

Alert Type Phishing

Severity Medium

Direction Inbound

Email Details

Field Value

Subject Unusual Sign-In Activity on Your Microsoft Account

Sender no-reply@microsoftsupport.co

Recipient c.allen@thetrydaily.thm

Analysis

The email attempted to impersonate Microsoft security services in order to trick the recipient into clicking a malicious login link. The attacker used fear and urgency tactics by claiming an unusual sign-in attempt had occurred on the user's Microsoft account.

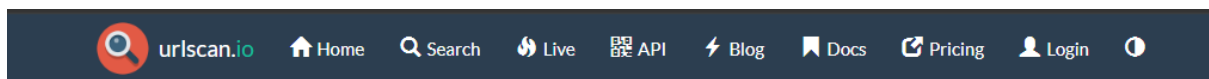
The message contained several phishing indicators:

- Microsoft impersonation attempt
- typosquatting domain (m1crosoftsupport.co)
- urgent security warning
- suspicious external login link
- credential harvesting attempt
- social engineering tactics

The embedded external login link directed users to:

<https://m1crosoftsupport.co/login>

The domain attempted to mimic Microsoft branding by replacing the letter "i" with the number "1", a common phishing technique used to deceive users into trusting malicious websites.



HTTP 400 Error

DNS Error - Could not resolve domain

Explanation

The domain m1crosoftsupport.co could not be resolved to a valid IPv4/IPv6 address. We won't try to load it in the browser.

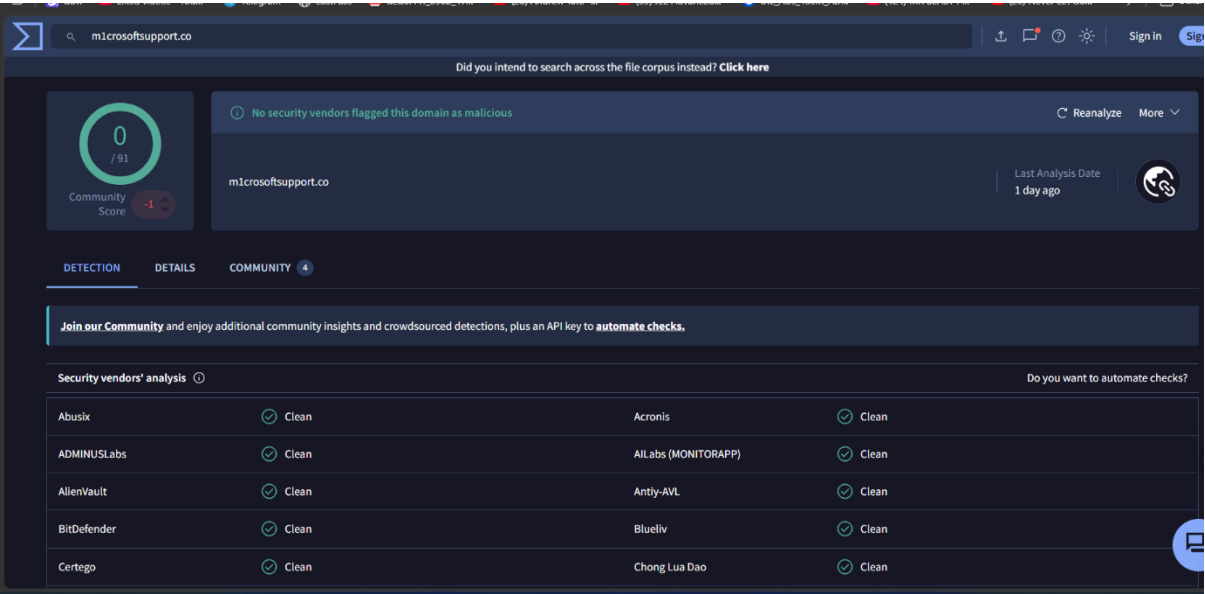
Domain Investigation

The domain m1crosoftsupport.co was investigated using VirusTotal and URLScan.

Findings included:

- VirusTotal did not flag the domain as malicious.
- URLScan returned a DNS resolution failure.
- The domain could not resolve to a valid IPv4/IPv6 address.
- The infrastructure appeared suspicious and non-legitimate.

Although VirusTotal showed no active detections, newly created or low-reputation phishing domains are not always immediately identified by security vendors. The typosquatting behavior and suspicious infrastructure strongly indicated phishing activity.



Risk Classification

Category Result

Threat Type Credential Harvesting Phishing

Risk Level Medium

INCIDENT REPORT

Time of Activity:

05/26/2026 14:23:12 UTC

List of Affected Entities:

- Recipient: c.allen@thetrydaily.thm
 - Suspicious Sender: no-reply@m1crosoftsupport.co
 - Suspicious Domain: m1crosoftsupport.co
 - Email Subject: Unusual Sign-In Activity on Your Microsoft Account
-

Reason for Classifying as True Positive:

The alert was classified as a True Positive phishing incident after investigation identified multiple phishing and impersonation indicators. The sender domain m1crosoftsupport.co attempts to mimic the legitimate Microsoft domain using typosquatting techniques by replacing the letter "i" with the number "1". The email used fear and urgency tactics by claiming an unusual sign-in attempt from Nigeria and urging the user to secure the account immediately. The embedded external login URL appeared designed to harvest user credentials. URLScan investigation showed that the domain could not resolve to a valid IP address, indicating suspicious or illegitimate infrastructure commonly associated with phishing campaigns.

Reason for Escalating the Alert:

The alert was escalated because the email impersonated Microsoft security services and attempted to lure the user into interacting with a suspicious login page. This activity presented a potential credential theft risk and could lead to unauthorized account access if the user interacted with the phishing link.

Recommended Remediation Actions:

- Quarantine and remove the phishing email from all affected mailboxes.
- Block the domain m1crosoftsupport.co across email and web security controls.
- Search the environment for similar phishing emails or related indicators.
- Notify the affected user about the phishing attempt.

- Advise users to avoid interacting with suspicious login or security alert emails.
 - Reset credentials immediately if the phishing link was accessed.
 - Add the identified indicators to SIEM detection rules and monitoring systems.
-

List of Attack Indicators:

- Suspicious sender address: no-reply@m1crosoftsupport.co
- Typosquatting domain: m1crosoftsupport.co
- Fake Microsoft security impersonation
- Suspicious external login link
- Urgent/fear-based social engineering language
- Claim of unusual sign-in activity from foreign location
- DNS resolution failure identified in URLScan
- Potential credential harvesting attempt

11. Common Phishing Indicators

During the investigation, several phishing patterns were consistently identified across different email samples.

Indicator	Description
Suspicious Domains	Unknown or spoofed sender domains
Urgent Language	Pressure tactics encouraging immediate action
Generic Greetings	Lack of personalization
Requests for Sensitive Information	Banking details, passwords, OTPs
Suspicious URLs	Links redirecting to fake websites
Unexpected Attachments	Potential malware delivery
Emotional Manipulation	Fear, rewards, curiosity, authority

These indicators are commonly observed in real-world phishing campaigns.

12. MITRE ATT&CK Mapping

The analyzed phishing activities align with several techniques from the [MITRE ATT&CK Framework](#).

Technique ID	Technique
T1566	Phishing
T1566.001	Spearphishing Attachment
T1566.002	Spearphishing Link
T1204	User Execution
T1059	Command and Scripting Interpreter

Mapping phishing attacks to MITRE ATT&CK techniques helps security analysts understand attacker behavior and improve detection strategies.

13. Prevention & Awareness Guidelines

Organizations should educate users on how to identify and report phishing attempts.

Best Practices

Do:

- Verify sender email addresses carefully
- Hover over links before clicking
- Use Multi-Factor Authentication (MFA)
- Report suspicious emails immediately
- Verify urgent requests through official channels

Don't:

- Share passwords or OTPs via email
- Open unknown attachments
- Trust unrealistic financial offers
- Click suspicious links
- Respond to suspicious external emails

Employee awareness significantly reduces the success rate of phishing attacks.

14. Recommendations

Based on the investigation findings, the following security recommendations are advised:

- Implement advanced email filtering solutions
- Conduct regular phishing awareness training
- Enable Multi-Factor Authentication (MFA)
- Monitor suspicious domains and indicators
- Perform simulated phishing exercises
- Strengthen endpoint protection systems
- Enforce email authentication protocols such as SPF, DKIM, and DMARC

These measures can help reduce phishing-related risks within organizations.

15. Conclusion

This project demonstrated practical phishing investigation techniques used in cybersecurity operations and SOC environments.

Multiple phishing email samples were analyzed to identify:

- suspicious domains
- social engineering tactics
- malicious indicators
- phishing techniques

The investigation highlighted how attackers manipulate users using urgency, financial rewards, fake branding, and malicious links.

The project also emphasized the importance of cybersecurity awareness and proactive threat detection in preventing phishing attacks.

Practical phishing analysis improves an organization's ability to identify threats early and reduce the likelihood of successful cyberattacks.

16. References

- [TryHackMe](#)
- [VirusTotal](#)
- [MXToolbox Email Header Analyzer](#)
- [Google Message Header Analyzer](#)
- [MITRE ATT&CK Framework](#)
- [Future Interns](#)